INCH-POUND

MIL-C-70576B (AR)
17 August 1993
See 6.6

MILITARY SPECIFICATION

CIRCUIT ASSEMBLY, BUILT-IN-TEST DISTRIBUTION BOX (PIVADS): 9360925

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Distribution Box Built-In-Test (BIT) circuit card assembly, herein after referred to as the Distribution Box BIT Board (DBB). The DBB circuits are used to communicate cannon control voltages and mount hatch protection circuits data to the Fire Control Processor (FCP). Data transmission is non-continuous and reception is serially formatted through a Universal Asynchronous Receiver Transmitter (UART) link which is enabled by the FCP Central Processing Unit (CPU) . DBB contains twelve microcircuits and resistive, capacitive, and semi-conductive circuitry. The DBB is a 4 by 8 1/2 inch double-layered circuit card mounted vertically in the Distribution Box. Refer to Figure 1 for the Distribution Box BIT assembly functional block diagram (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document **or by letter.**

AMSC N/A **FSC 1285**

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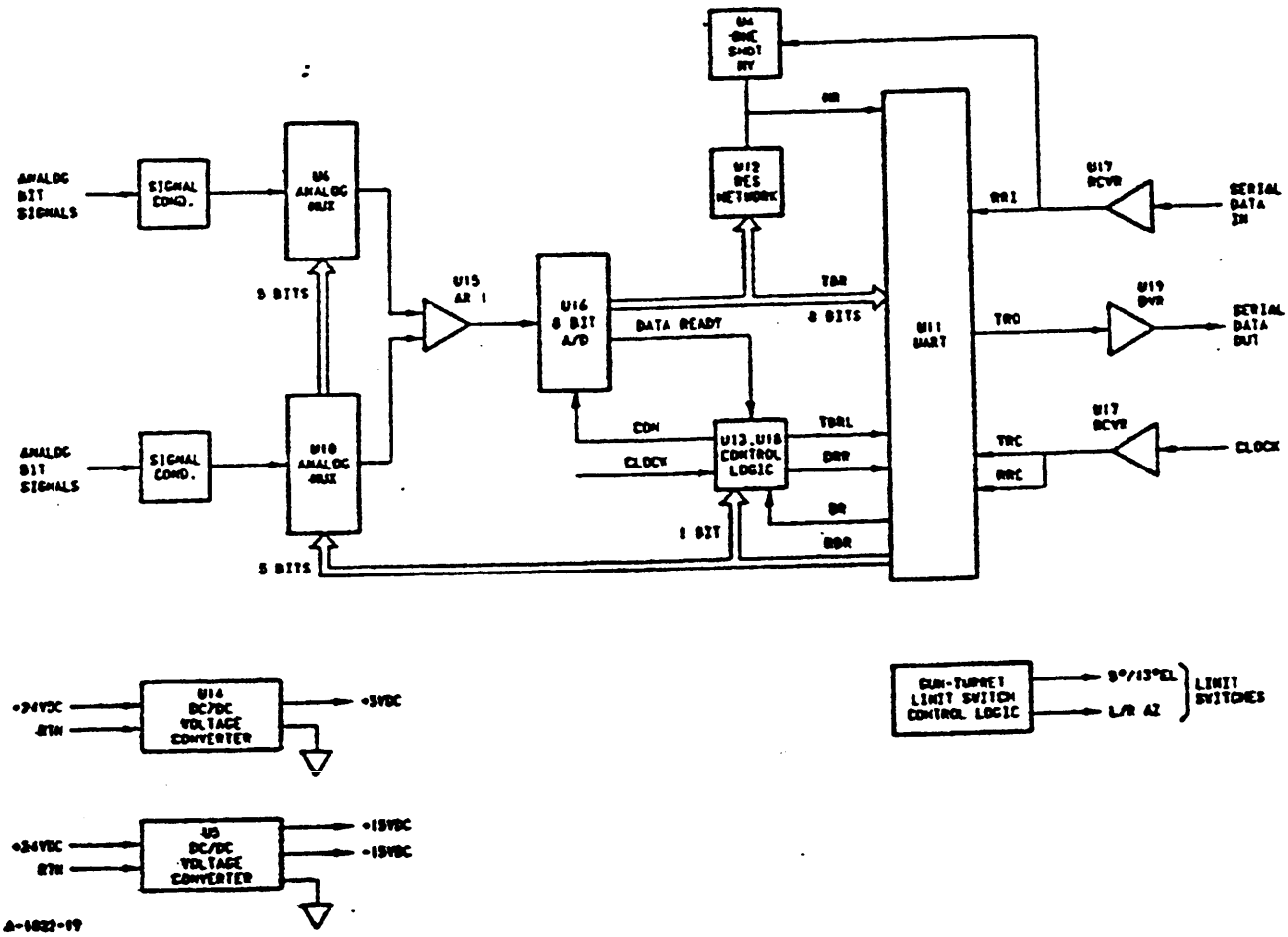


FIGURE 1. Distribution Box BIT Assembly Functional Block Diagram

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS (see 6.4)

Us. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360925	-	BIT Board Assembly, Distribution Box
9360927	-	BIT Board, Distribution Box, Schematic Diagram
12950877	-	Requirements for Soldering Electrical and Electronic Assemblies

TEST SET

TPS 9361342	-	Test Program Set Manual
TPT 9361342	-	Test Program Set Tape

SPECIAL PACKAGING INSTRUCTION

PS 9360925	-	Packaging of Distribution Box BIT Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

IPC-DW-425	-	Design and End Product Requirements for Discrete Wiring Boards
EIA 422	-	Electrical Characteristics of Balanced Voltage Digital Interface Circuits

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however . . . supersedes applicable laws and regulations unless a specific exemption has been "obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comly with all requirements specified on Drawing 9360925, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the DBB assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.2.1 IPC-DW-425. IPC-DW-425/11 Type Cl plated through connection.

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3.3 Environmental service conditions.

3.3.1 Vibration The DBB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the eqivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The DBB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's * 10 percent, 11 milliseconds ± 10 percent, half-sine wave applied in each direction of three mutually perpendicular axes.

3.3.3 Altitude operating. The DBB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg) .

3.3.4 Temperature.

3.3.4.1 Operating. The DBB shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F (equipment used in a sheltered environment) , -40T and +155°F (equipment used in a field environment).

3.3.4.2 Non-onerating (storage) . The DBB shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160"F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The DBB shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the DBB.

3.5 Performance.

3.5.1 Functional description. The DBB circuit shall receive an address in serial format, and select a particular analog voltage which shall be converted to digital data and transmitted serially. The UART has two parts: a receiver and a transmitter. The receiver shall take a serial formatted byte and

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convert it into a parallel data byte and output it at the Transmitter Buffer Register (TBR). This received byte shall be used to address the two multiplexer for the selection of a particular analog signal at the inputs of the multiplexer. 'Ibis analog signal shall be converted into a digital signal by an analog to digital (A/D) converter. The parallel data shall then be loaded into the UART for transmission. The serial signals shall be buffered by differential line drivers and receivers. The serial signal at the output of the receiving buffer shall be input into the UART at a clock rate of **31.25** KHz. when the entire serial byte has been received, the data byte shall appear in parallel form, tied to the address select lines at the two multiplexer. The A/D converter shall convert the analog signal at its input. This analog signal shall be selected by the data byte, and the digital signal shall appear at the output of the A/D converter. The UART shall load the digitalized data byte for serial transmission. This transmitter function shall be executed only if the most significant bit (MSB) of the received byte indicates the need for this data. The UART shall be master reset by a one-shot multivibrator approximately 68 msec after the last down edge of the last byte received at the Serial Data Input. All input signals shall be scaled so that the A/D converter shall always ***receive*** a normalized input in the ± 5 V range. The scaling shall be done by a voltage divider network before the signals reach the multiplexer. To match the Transistor-Transistor Logic (TTL) level of the circuit, a pull-up resistor array shall be used at the UART to pull up the Complementary Metal Oxide Semi-Conductor (CMOS) logic level. The DBB circuit shall include steering diodes for the limit switch functions. The steering diodes shall have no internal connection with the rest of the DBB circuit.

3.5.2 Power requirements. The DBB power shall be input at connector P1-20, 21 = +24 V BAT and PI-55, 56 = **+24 V BAT RTN** with requirements as follows:

	MIN	TYP	MAX	UNITS
Input Voltage	16.0	24.0	32.0	Vdc
Input Current	344	458	687	mAdc
Input Impedance	23.3	52.4	93.2	ohms
Input Power	---	11	---	Watts

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With the above input, the on-board dc to dc converter outputs at the test points shall be as follows:

Test Point	Voltage (Vdc)	Ripple (mVrms)
PI-22	+5.00 ± 0.25	80 max.
PI-23	+15.00 ± 0.25	40 max.
PI-57	-15.00 ± 0.25	40 max.

3.5.3 Clock requirements. The DBB clock (500 Hz) signal applied at PI-1 (HI) and PI-36 (LO) shall be differential with RS-422 voltage levels per 3.5.13. The clock frequency shall be 500 KHz ± 74 Hz.

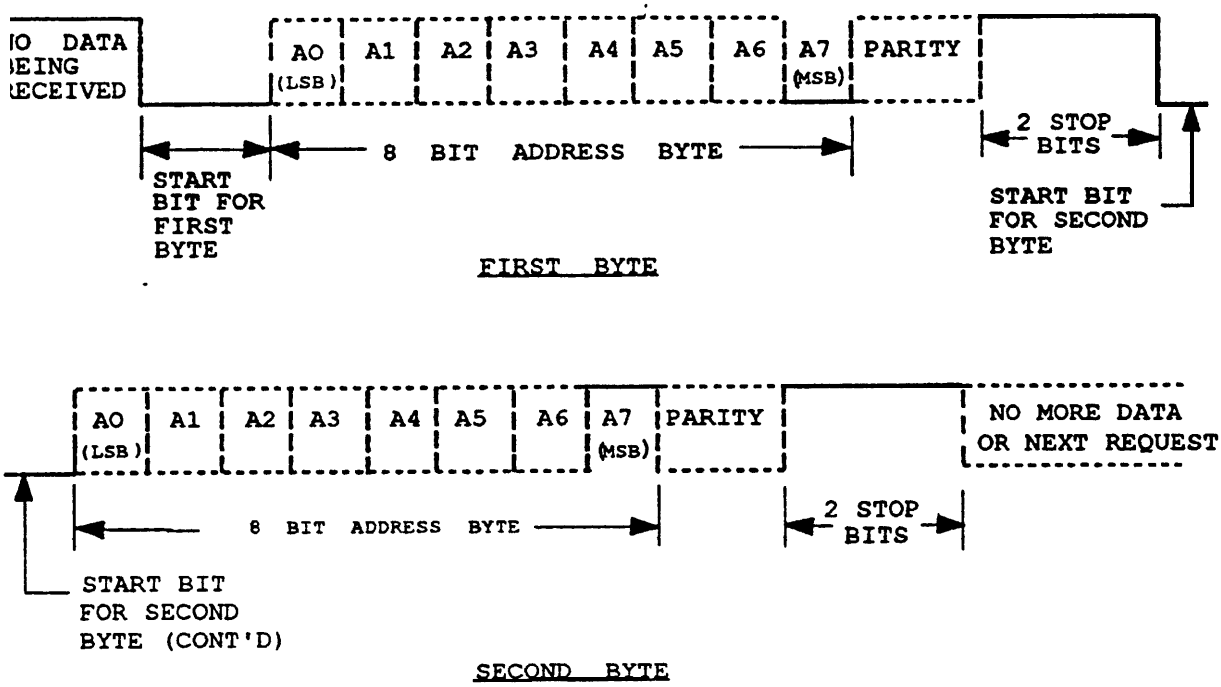
3.5.4 Format of UART receiver address bytes. TWO address bytes must be received by the DBB in order to transmit out the digital byte which represents the requested multiplexer analog input voltage. The first byte has the A7 bit set low to load the UART'S Receiver Buffer Register. The second byte has the A7 bit set high to cause the low-to-high transition of TBRL, which transfers the data in the UART'S Transmitter Buffer Register to the Transmitter Register where it is serially clocked out. The address byte input into the DBB shall be differential signals with RS-422 voltage levels per 3.5.13 applied at P1-2 (HI) and P1-37 (Lo). With power applied to the DBB per 3.5.2, the address data shall have TTL voltage levels per 3.5.11 at P1-58 (RRI) test point except that a pull up resistor shall ensure the minimum 3.0 V CMOS threshold required by the UART. The format for the address words shall be per Figure 2.

3.5.5 Format of UART transmitter outputs. Each byte transmitted from the DBB shall be differential with RS-422 voltage levels per 3.5.13, output at PI-3 (HI) and P1-38 (LO). The transmitter output byte format shall be per Figure 3.

3.5.6 Master Reset (MR) functions. For the following requirements, P1-47 must be connected to P1-48.

3.5.6.1 UART not receiving address data. Upon applying power per 3.5.2 and lock input per 3.5.3 to the DBB, when no address data is being clocked into the UART, RESET OUT (P1-47) and RESET IN (PI-48) shall go high to enable the MR function of the UART. When enabled, the UART'S Data Received (DR) line (Pl-50) shall go low and Transmitter Register Output (TRO) line shall go high to initialize the DBB for receiving data and inhibiting data transmission.

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NOTES:

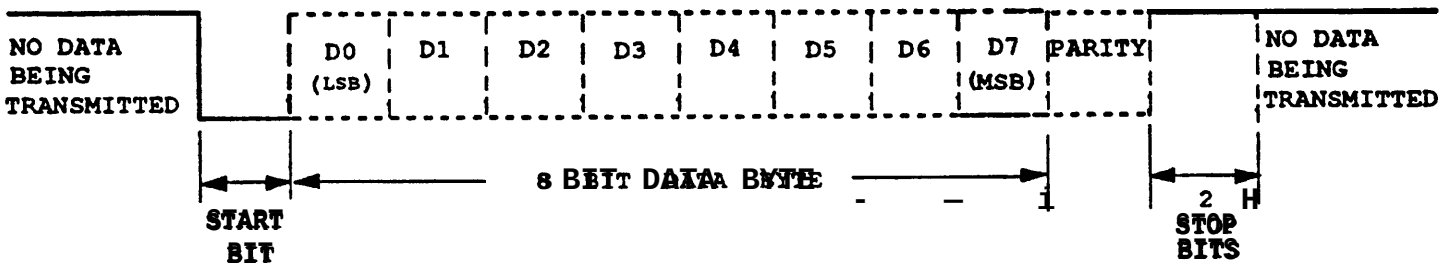
1. The data line is always high when no address bytes are being received.
2. Each start bit is always low and each stop bit Is always high.
3. The Least Significant Bit(LSB) Is the first address bit received in each byte. The Most Significant Bit(LSB) is the last address bit received in each byte.
4. When an address byte has been received by the UART, it is output on the UART'S Receiver Buffer Register (RBR) lines as follows:

   Bit AO is on RBR1
   Bit Al is on RBR2
   .
   Bit A7 is on RBR8

5. Address bits A5 and A6 are "don't cares" and may be high or low, because RBR lines RBR6 and RBR7 are not connected.
6. All bits of the second address byte are identical to those of the first address byte except for A7 and parity".

FIGURE 2. Format of UART receiver address bytes.

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NOTES :

1. The data line is always high when no data is being transmitted.

2. The start bit is always low and the 2 stop bits are always high.

3. The Least Significant Bit (LSB) is the first data bit transmitted in each data byte. The Most Significant Bit (MSB) is the last data bit transmitted in each byte.

4. The parity bit is odd; the UART sums data bits DO through D7 and adds a one (high parity bit) or adds a zero (low parity bit) to cause the sum plus parity bit to be an odd number.

5. Each data bit for transmission is input into the UART on the UART'S Transmitter Buffer Register (TBR) lines as follows:

   Bit DO is on TBR1
   Bit D1 is on TBR2

   Bit D7 is on TBR8

FIGURE 3. Format Of UART transmitter output bytes.

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3.5.6.2 UART receiving address data. With power per 3.5.2 and clock input per 3.5.3 applied to the DBB, when serial address data is being clocked into the UART at CPU TO DISTRIBUTION BOX DATA with P1-2 (HI) and P1-37 (LO), RESET OUT (P1-47) and RESET IN (P1-48) shall go low and disable the MR function of the UART to allow the DR and TRO lines to toggle. The low shall be for a time delay triggered by the high to low transition of the first start bit and extended by every high to low transition (retrigger) within each received byte. Each trigger and re- trigger shall begin a 60 ± 10 millisecond delay (noncumlative)t which shall be sufficient to keep the MR line low while address data is being clocked into the UART. After the delay, RESET OUT (P1-47), RESET IN (P1-48) shall be high to again initialize the DBB . DR (P1-50) shall already be low due to DRR going low on the second leading edge of the 500 kHZ clock after completion of a data conversion.

3.5.7 Analog multiplexer, A/D converter and UART transmitter circuit. Signal timing shall be as shown in Figure 4.

3.5.7.1 Initialization. Upon applying power per 3.5.2 and clock input per 3.5.3, the DBB shall initialize (reset) Per 3.5.6.1.

3.5.7.2 Multiplexer U6 channels. Per Tables I (U6 Channels. Connector J1 and buffer amplifier inputs) and 11 (U6 Channel. A/D inputs/outputs and transmitted outputs) simul- taneously applying each Vin to its respective nominal analog input pin and clocking in the address byte for the DBB U6 multiplexer channel S1 per Table I shall cause the following outputs with values per Table I:

a. Received address bits A7, A4, A3, A2, Al and AO at P1-40, 41, 42, 43, 44, 45, 46 respectively, and,

b. Buffer input voltage IN1 and IN2 at P1-58.

Clocking in the second address byte per Table II for U6 multiplexer channel S1 shall cause the following outputs with values per Table II:

(1) A/D input voltage (A/D IN) at pin P1-49

(2) A/D output byte D7 through DO at P1-5 through 12, respectively

(3) DISTRIBUTION BOX TO CPU DATA at P1-3 (HI) and Pl-38 (LO) with format per Figure 3.

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d. Similarly, the remaining channels of multiplexer U6 shall have all of its outputs as specified in Tables I and II.

3.5.7.3 Multiplexer U10 channels. With the analog input of 3.5.7.2 applied, and with the U10 multiplexer channel addresses sequenced per Tables III and IV, each U12 channel shall have all of its outputs as specified in Table III (U10 Charnels. Connector P1 and buffer amplifier inputs) and Table IV (U10 Channels. A/D inputs/outputs and transmitted outputs)=

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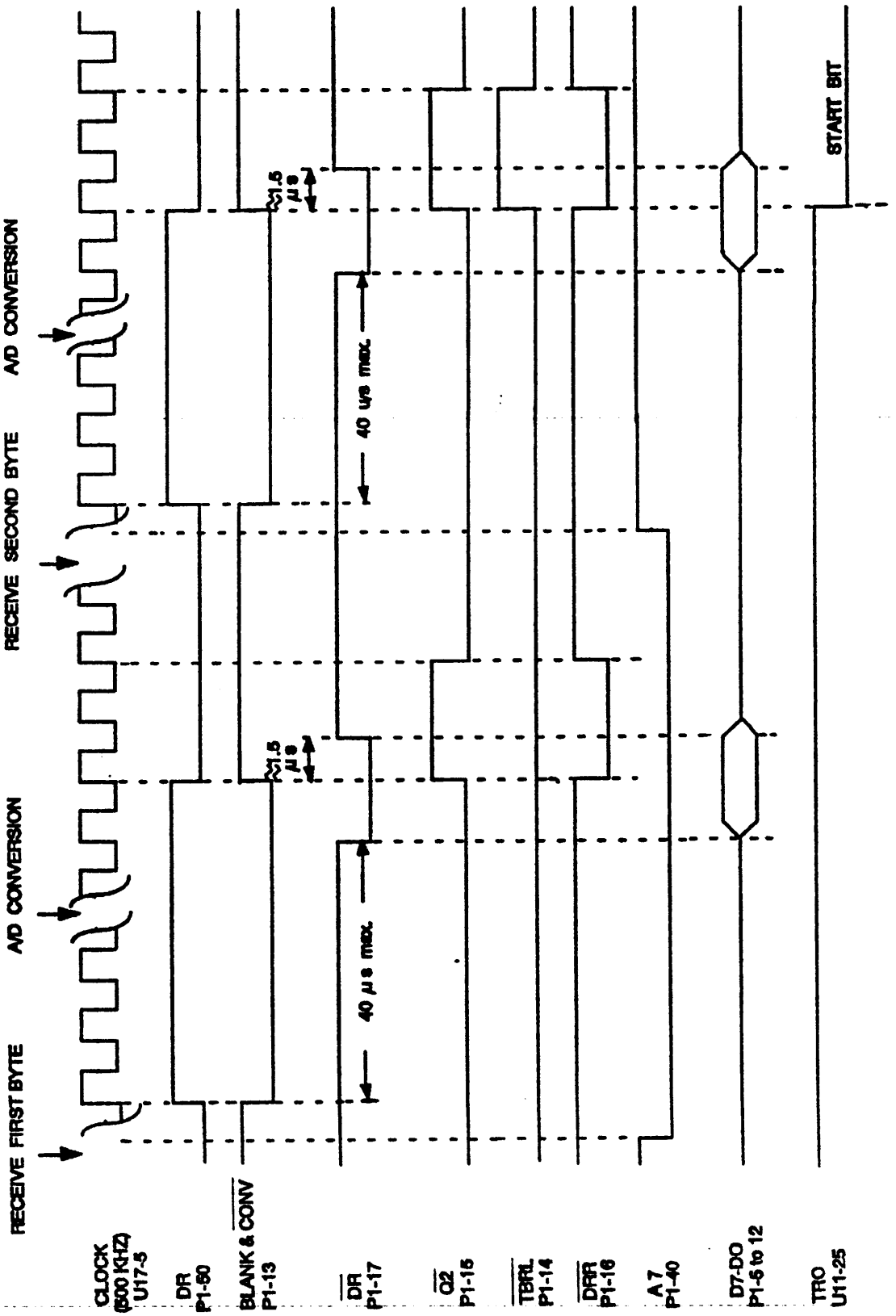


FIGURE 4. Multiplexer, converter and UART transmitter timing diagram.

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TABLE I. U6 channels. Connector P1 and buffer amplifier inputs.

U6 MUX CHANNEL	MUX ADDRESS A7 A6 A5 A4 A3 A2 A1 A0	$\overline{A4}$	CONN. PIN NO.	INPUT DESCRIPTION	Vin (Vdc unless stated otherwise)	BUFFER INPUT VOLTAGE IN1 AND IN2 (Vdc)
S1	1 X X 0 0 0 0 0	1	INTERNAL	+15 V TEST	+14.750 TO +15.250	+3.605 TO +3.880
S2	1 X X 0 0 0 0 1	1	P1-34	GUN MOTOR DRIVE	+23.990 TO +24.010	+3.335 TO +3.481
# S3	1 X X 0 0 0 1 0	1	P1-34	GUN MOTOR DRIVE SAFETY	OPEN	+1.580 TO +2.580
# S3	1 X X 0 0 0 1 0	1	P1-34	GUN MOTOR DRIVE SAFETY	+23.990 TO +24.010	+4.570 TO +4.820
S4	1 X X 0 0 0 1 1	1	P1-33	BURST LIMIT RELAY	+23.990 TO +24.010	+3.335 TO +3.481
S5	1 X X 0 0 1 0 0	1	P1-32	DRUM EMPTY	+23.990 TO +24.010	+3.335 TO +3.481
S6	1 X X 0 0 1 0 1	1	P1-31	GUN BRAKE	+23.990 TO +24.010	+3.335 TO +3.481
# S7	1 X X 0 0 1 1 0	1	P1-31	GUN BRAKE SAFETY	OPEN	+1.580 TO +2.580
# S7	1 X X 0 0 1 1 0	1	P1-31	GUN BRAKE SAFETY	+23.990 TO +24.010	+4.570 TO +4,820
S8	1 X X 0 0 1 1 1	1	P1-30	CLEAR RELAY	+23.990 TO +24.010	+3.335 TO +3.481
# S9	1 X X 0 1 0 0 0	1	P1-30	CLEAR RELAY SAFETY	OPEN	+1.580 TO +2.580
# S9	1 X X 0 1 0 0 0	1	P1-30	CLEAR RELAY SAFETY	+23.990 TO +24.010	+4.570 TO +4.820
S10	1 X X 0 1 0 0 1	1	P1-25	FIRING VOLTS 1 (370 V)	+369.00 TO +371.00	+4.760 TO .4.971
S11	1 X X 0 1 0 1 0	1	INTERNAL	BAT	+23.990 TO +24.010	+3.335 TO +3.481
S12	1 X X 0 1 0 1 1	1	P1-69	HIGH VOLTAGE GUN BAT	+23.990 TO +24.010	+3.335 TO +3.481
S13	1 X X 0 1 1 0 0	1	P1-68	TIME DELAY	+23.990 TO +24.010	+3.335 TO +3.481
S14	1 X X 0 1 1 0 1	1	P1-60	FIRING VOLTS 2 (370 V)	+369.00 TO +371.00	+3.606 TO +3.774
S15	1 X X 0 1 1 1 0	1	P1-67	CLEAR RELAY COIL	+23.990 TO +24.010	+3.335 TO +3.481
S16	1 X X 0 1 1 1 1	1	INTERNAL	-15 V TEST	-14.750 TO -15.250	-3.605 TO -3.880

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TABLE II. U6 channels. A/D inputs/outputs and transmitted outputs.

U6 MUX CHANNEL	MUX ADDRESS A7	A6	A5	A4	A3	A2	A1	A0	A/D INPUT VOLTAGE A/D IN (Vdc)	A/D, DISTRIBUTION BOX FROM D7	D6	D5	D4	D3	D2	D1	D0	TO CPU OUTPUT WORD TO D7	D6	D5	D4	D3	D2	D1	D0
S1	1	X	X	0	0	0	0	0	+3.595 TO +3.880	1	1	0	1	1	0	1	1	1	1	1	0	0	1	0	0
S2	1	X	X	0	0	0	0	1	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
# S3	1	X	X	0	0	0	1	0	+1.575 TO +2.585	1	0	1	0	1	0	0	0	1	1	0	0	0	0	1	0
# S3	1	X	X	0	0	0	1	0	+4.565 TO +4.825	1	1	1	1	0	1	0	0	1	1	1	1	1	1	0	0
S4	1	X	X	0	0	0	1	1	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S5	1	X	X	0	0	1	0	0	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S6	1	X	X	0	0	1	0	1	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
# S7	1	X	X	0	0	1	1	0	+1.575 TO +2.585	1	0	1	0	1	0	0	0	1	1	0	0	0	0	1	0
# S7	1	X	X	0	0	1	1	0	+4.565 TO +4.825	1	1	1	1	0	1	0	0	1	1	1	1	1	1	0	0
S8	1	X	X	0	0	1	1	1	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
# S9	1	X	X	0	1	0	0	0	+1.575 TO +2.585	1	0	1	0	1	0	0	0	1	1	0	0	0	0	1	0
# S9	1	X	X	0	1	0	0	0	+4.565 TO +4.825	1	1	1	1	0	1	0	0	1	1	1	1	1	1	0	0
S10	1	X	X	0	1	0	0	1	+4.750 TO +4.971	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	1
S11	1	X	X	0	1	0	1	0	+3.325 TO +3.461	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S12	1	X	X	0	1	0	1	1	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S13	1	X	X	0	1	1	0	0	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S14	1	X	X	0	1	1	0	1	+3.596 TO +3.774	1	1	0	1	1	0	1	1	1	1	1	0	0	0	1	0
S15	1	X	X	0	1	1	1	0	+3.325 TO +3.481	1	1	0	1	0	1	0	0	1	1	0	1	1	0	1	0
S16	1	X	X	0	1	1	1	1	-3.595 TO -3.880	0	0	1	0	0	1	0	1	0	0	0	1	1	0	1	1

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TABLE III. U10 channels. Connector P1 and buffer amplifier inputs.

U10 MUX CHANNEL	MUX ADDRESS A7 A6 A5 A4 A3 A2 A1 A0	$\overline{A4}$	CONN. PIN NO.	INPUT DESCRIPTION	Vin (Vdc unless stated otherwise)	BUFFER INPUT VOLTAGE IN1 AND IN2 (Vdc)
S1	0 X X 1 0 0 0 0	0	INTERNAL	-15 V TEST	-14.750 TO -15.250	-3.605 TO -3.880
S2	0 X X 1 0 0 0 1	0	P1-28	TIME DELAY RELAY COIL	+23.990 TO +24.010	+3.335 TO +3.481
S3	0 X X 1 0 0 1 0	0	P1-27	RADAR POWER	+23.990 TO +24.010	+3.335 TO +3.481
S4	0 X X 1 0 0 1 1	0	P1-70	TRIGGER RELAY	+23.990 TO +24.010	+3.335 TO +3.481
S5	0 X X 1 0 1 0 0	0	P1-63	AZ CW LIMIT	+23.990 TO +24.010	+3.335 TO +3.481
S6	0 X X 1 0 1 0 1	0	P1-64	AZ CCW LIMIT	+23.990 TO +24.010	+3.335 TO +3.481
S7	0 X X 1 0 1 1 0	0	P1-65	UP LIMIT	+23.990 TO +24.010	+3.335 TO +3.481
S8	0 X X 1 0 1 1 1	0	P1-66	DOWN LIMIT	+23.990 TO +24.010	+3.335 TO +3.481
S9	0 X X 1 1 0 0 0	0	P1-29	FIRE RELAY COIL	+23.990 TO +24.010	+3.335 TO +3.481
S10	0 X X 1 1 0 0 1	0	P1-35	LOW VOLTAGE GUN BAT	+ 9.990 TO +10.010	+3.264 TO +3.404
S11	0 X X 1 1 0 1 0	0	INTERNAL	+5 V	+ 4.750 TO + 5.250	+4.750 TO +5.250
S12	0 X X 1 1 0 1 1	0	INTERNAL	GROUND	- 0.010 TO + 0.010	-0.010 TO +0.010
S13	0 X X 1 1 1 0 0	0	NO	CONNECTION	--- ---	--- ---
S14	0 X X 1 1 1 0 1	0	NO	CONNECTION	--- ---	--- ---
S15	0 X X 1 1 1 1 0	0	NO	CONNECTION	--- ---	--- ---
S16	0 X X 1 1 1 1 1	0	INTERNAL	+15 v	+14.750 TO +15.250	+3.605 TO +3.880

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TABLE IV. U10 channels. A/D inputs/outputs and transmitted outputs.

U10 MUX CHANNEL	MUX ADDRESS A7 A6 A5 A4 A3 A2 A1 A0	A/D INPUT VOLTAGE A/D IN (Vdc)	A/D, DISTRIBUTION BOX FROM D7 D6 D5 D4 D3 D2 D1 D0	TO CPU OUTPUT WORD TO D7 D6 D5 D4 D3 D2 D1 D0
S1	1 X X 1 0 0 0 0	+3.595 TO +3.880	0 0 1 0 0 1 0 1	0 0 0 1 1 0 1 1
S2	1 X X 1 0 0 0 1	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S3	1 X X 1 0 0 1 0	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S4	1 X X 1 0 0 1 1	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S5	1 X X 1 0 1 0 0	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S6	1 X X 1 0 1 0 1	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S7	1 X X 1 0 1 1 0	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S8	1 X X 1 0 1 1 1	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S9	1 X X 1 1 0 0 0	+3.325 TO +3.481	1 1 0 1 0 1 0 0	1 1 0 1 1 0 1 0
S10	1 X X 1 1 0 0 1	+3.254 TO +3.404	1 1 0 1 0 0 1 0	1 1 0 1 1 0 0 0
S11	1 X X 1 1 0 1 0	+4.740 TO +5.250	1 1 1 1 1 0 0 0	1 1 1 1 1 1 1 1
S12	1 X X 1 1 0 1 1	-0.020 TO +0.010	0 1 1 1 1 1 1 1	1 0 0 0 0 0 0 0
S13	1 X X 1 1 1 0 0	NO CONNECTION	- - - - - - - -	- - - - - - - -
S14	1 X X 1 1 1 0 1	NO CONNECTION	- - - - - - - -	- - - - - - - -
S15	1 X X 1 1 1 1 0	NO CONNECTION	- - - - - - - -	- - - - - - - -
S16	1 X X 1 1 1 1 1	+3.595 TO +3.880	1 1 0 1 1 0 1 1	1 1 1 0 0 1 0 1

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3.5.8 **Pull-up resistors.** The following signals shall be maintained at a TTL logic high per 3.5.11 by pull-up resistors when the DBB is powered up per 3.5.2:

a. P1-4 U18-SET

b. P1-39 U18-RESET

c. P1-51 TEST TBRL

3.5.9 Diode network. The DBB shall contain a diode network (D5-D12) which hall include steering diodes for the azimuth and elevation mount limit switches and kickback diodes for limit switch relays as specified in Tables V (Steering diodes) and VI (Kickback diodes).

TABLE V. Steering diodes.

corm. Pin	Forward Bias Diode	Corm.
P1-18	D6	P1-62
P1-18	D7	P1-54
P1-18	Dll	P1-52
P1-19	D8	P1-54
P1-19	D12	P1-52

TABLE VI. Kickback diodes.

From Conn. Pin	Through Reverse Bias Diode	To Ground Through
P1-51	D5	Rll
P1-53	D10	R10
P1-54	D9	R9

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3.5.10 Input/Output requirements. The DBB shall have inputs as specified in Table VII (DBB inputs) and outputs as specified in Table VIII (DBB outputs).'

3.5.11 TTL voltage levels. The TTL voltage levels for inputs and outputs are as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	0.2	0.4
Voltage Output High (VOH)	2.4	3.4	---
Voltage Input Low (VIL)	----	--	0.7
Voltage Input High (VIH)	2.0	---	---

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TABLE VII. DBB inputs.

CONNECTOR PIN	SIGNAL NAME	DESIGNATION	REQUIREMENTS
P1-1	DISTRIBUTION BOX CLOCK (500 KHz), HI	U17-6	RS-422 PER 3.5.13
P1-2	CPU TO DISTRIBUTION BOX DATA, LO	U17-2	RS-422 PER 3.5.13
P1-18	GUN POSITION SWITCH	DIODE NETWORK	ANALOG INPUT, 0-24 V
P1-19	ELEVATION -5 DEG. LIMIT	DIODE NETWORK	ANALOG INPUT, 0-24 V
P1-25	FIRING VOLTAGE 1 (370 V)	U6-10	ANALOG INPUT, 0-370 V
P1-27	RADAR POWER	U10-21	ANALOG INPUT, 0-24 V
P1-28	TIME DELAY RELAY COIL	U10-20	ANALOG INPUT, 0-24 V
P1-29	FIRE RELAY COIL	U10-11	ANALOG INPUT, 0-24 V
P1-30	CLEAR RELAY	U6-26	ANALOG INPUT, 0-24 V
P1-31	GUN BRAKE	U6-24	ANALOG INPUT, 0-24 V
P1-32	DRUM EMPTY	U6-23	ANALOG INPUT, 0-24 V
P1-33	BURST LIMIT RELAY	U6-22	ANALOG INPUT, 0-24 V
P1-34	GUN MOTOR DRIVE	U6-20	ANALOG INPUT, 0-24 V
P1-35	LOW VOLTAGE GUN BATTERY	U10-10	ANALOG INPUT, 0-24 V
P1-36	DISTRIBUTION BOX CLOCK (500 KHz), LO	U17-7	RS-422 PER 3.5.13
P1-37	CPU TO DISTRIBUTION BOX DATA, LO	U17-1	RS-422 PER 3.5.13
P1-48	RESET IN	U12-15, U-11-21	TTL PER 3.5.11
P1-60	FIRING VOLTAGE 2 (370 V)	U6-6	ANALOG INPUT, 0-370 V
P1-63	AZ CW LIMIT	U10-23	ANALOG INPUT, 0-24 V
P1-64	AZ CCW LIMIT	U10-24	ANALOG INPUT, 0-24 V
P1-65	UP LIMIT	U10-25	ANALOG INPUT, 0-24 V
P1-66	DWN LIMIT	U10-26	ANALOG INPUT, 0-24 V
P1-67	CLEAR RELAY COIL	U6-5	ANALOG INPUT, 0-24 V
P1-68	TIME DELAY	U6-7	ANALOG INPUT, 0-24 V
P1-69	HIGH VOLTAGE GUN BATTERY	U6-8	ANALOG INPUT, 24 V
P1-70	TRIGGER RELAY	U10-22	ANALOG INPUT, 0-24 V

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TABLE VIII. **DEB** outputs.

CONNECTOR PIN	SIGNAL NAME	**DESIGNATION**	**REQUIREMENTS**
P1-3	DISTRIBUTION BOX *TO* **CPU** DATA, HI	U19-2	RS-422 PER 3.5.13
P1-38	DISTRIBUTION BOX TO CPU DATA, LO	U19-3	RS-422 PER 3.5.13
P1-47	RESET OUT	U4-6	TTL PER 3.5.11
P1-62	EL-5 DEG. LIMIT ENABLE	DIODE NETWORK	0-24 V, 30 mA
P1 -52	AZ CW LIMIT RELAY COIL	DIODE NETWORK	0-24 V, 30 MA
P1-54	**AZ CCW LIMIT RELAY COIL**	**DIODE NETWORK**	**0-24 V, 30 mA**
P1-53	EL-5 DOWN **DEG LIMIT RELAY COIL**	**DIODE NETWORK**	**0-24 V, 30 mA**

3.5.12 CMOS voltage levels. The CMOS voltage levels for inputs and outputs are as follows:

	MIN	TYP	MAX
Voltage Output Low (VOL)	----	--	0.45
Voltage Output High (VOH)	2.4	---	---
Voltage Input Low (VIL)	----	--	0.8
Voltage Input High (VIH)	3.5	---	---

3.5.13 RS-422 inputs/outputs.

3.5.13.1 Differential voltages. The RS-422 clock, line receiver and line transmitter buffer differential voltage level shall be as follows:

	High logic level	Low logic level
Maximum input:	+5.5 v	**-5.5 v**
Sensitivity:	Vih = Vin-vin > +0.5 V	Vi 1 = **Vin-Vin < -0.5 V**

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3.5.13.2 common mode voltages. The Rs-422 common mode voltage levels shall be:

	MIN	TYP	MAX
Voltage Output Low (VOL)	---	---	0.45
Voltage Output High (VOH)	2.5	3.2	---
Voltage Input Low (VIL)	---	---	0.80
Voltage Input High (VIH)	2.0	---	---

3.5.13.3 Frequencies. The clock and receiver/transmitter data frequencies shall be:

clock - 500.000 kHz ± 74 Hz

data - 31.250 kHz ± 5 Hz

3.5.14 Test points. The DBB shall have **test points as** specified in Table IX (Test points) .

3.5.15 Guard Pins. Connector P1 shall include 4 guard pins to provide isolation of the firing voltage (370 Vdc). These pins shall not be utilized as spares. The guard pins shall be as follows:

P1-24	P1-59
P1-26	P1-61

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

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TABLE IX. Test points.

CONNECTOR PIN	SIGNAL NAME	DESIGNATION	REQUIREMENTS
P1-4	U18-SET	U18-4	TTL PER 3.5.11
P1-5	D7	U16-9	TTL PER 3.5.11
P1-6	D6	U16-8	TTL PER 3.5.11
P1-7	D5	U16-7	TTL PER 3.5.11
P1-8	D4	U16-6	TTL PER 3.5.11
P1-9	D3	U16-5	TTL PER 3.5.11
P1-10	D2	U16-4	TTL PER 3.5.11
P1-11	D1	U16-3	TTL PER 3.5.11
P1-12	D0	U16-2	TTL PER 3.5.11
P1-13	BLANK AND $\overline{\text{CONV}}$	U13-8	TTL PER 3.5.11
P1-14	$\overline{\text{TBRL}}$	U13-6	TTL PER 3.5.11
P1-15	Q^2	U18-8	TTL PER 3.5.11
P1-16	$\overline{\text{DRR}}$	U18-9	TTL PER 3.5.11
P1-17	$\overline{\text{DR}}$	U16-17	TTL PER 3.5.11
P1-22	+5 VOLTS	U14-12	+5 PER 3.5.2
P1-23	+15 VOLTS	U5-10	+15 PER 3.5.2
P1-39	U18-RESET	U18-1	TTL PER 3.5.11
P1-40	A7	U11-5	CMOS PER 3.5.12
P1-41	$\overline{\text{A4}}$	U13-11	TTL PER 3.5.11
P1-42	A4	U11-8	CMOS PER 3.5.12
P1-43	A3	U11-9	CMOS PER 3.5.12
P1-44	A2	U11-10	CMOS PER 3.5.12
P1-45	A1	U11-11	CMOS PER 3.5.12
P1-46	A0	U11-12	CMOS PER 3.5.12
P1-49	A/D IN	AR1-1	ANALOG -5 TO +5
P1-50	DR	U11-19	CMOS PER 3.5.12
P1-51	TEST TBRL	U13-5	TTL PER 3.5.11
P1-57	-15 VOLTS	U5-12	-15 V PER 3.5.2
P1-58	IN 1 AND IN 2	AR1-3	ANALOG -5 V TO +5 V

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification **shall become a part of the contractor's** overall **inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.**

4.1.2 . General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling **inspection (see 4.5) .**

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	**90% RH Max.**
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Distribution Box Bit Board	9360925	3

4.3.2 Inspections to be performed. ***As*** detemined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to

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exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP) . In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., ***may*** be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

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4.4.2.3 Functional tests. The tests in Table X shall be performed on a 100 percent basis.

TABLE X. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical	None	N/A
Major 101. Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

4.5 Special sampling. One DBB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at **random. The sample shall have met the requirements and tests in Table X and shall then meet the requirements and tests in Table** XI. **Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.**

TABLE XI. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical	None	N/A
Major 103. vibration 104. Shock 105. Altitude 106. Temperature 107. Humidity	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

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4.5.1 Failure of sample. Should" any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government desigm. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor desire. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

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4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table XII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE XII. Test equipment list.

ITEM NUMBER	DESCRIPTION
1. Interface Device	9397373
2. Test Plug, P14	9397559
3. Digital Card Tester	AN/USM-465A
4. Comcode Program Tape	Version 2.0
5. Test Program Set Manual	TPS9361342
6. Test Program Set Tape	TPT9361342

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I, only steps 4 through 8 of paragraph 11-2.1 apply. The DCP BIT assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the DCP BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10 percent amplitude and 11 ms ± 10 percent duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10 percent.

d. The shock measuring system shall have a flat frequency response within ± 10 percent from 20 to 2,000 hertz.

After the application of all the shock impulses, the DCP BIT assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedure II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2, 4.7.3, for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

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4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Format of UART receiver address bytes. The UART receiver address shall perform as per 3.5.4 and shall be tested in accordance with TPS9361342.

4.7.3.2 Format of UART transmitter outputs. The UART transmitter output shall perform as per 3.5.5 and shall be tested in accordance with TPS9361342.

4.7.3.3 Master reset (MR) function. The master reset function shall perform as per 3.5.6 and shall be tested in accordance with TPS9361342.

4.7.3.4 Analog multiplexer, A/D converter and UART transmitter circuit. The analog MUX, A/D converter and UART transmitter shall perform as per 3.5.7 and shall be tested in accordance with TPS9361342.

4.7.3.5 Power requirements. The power requirements shall perform as per 3.5.2 and shall be tested in accordance with TPS9361342.

4.7.3.6 Input/outmut requirements. The input/output requirements perform as per 3.5.10 and shall be tested in accordance with TPS9361342.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph **3.7.**

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360925. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The DBB is used in the Distribution Box of the PIVADS self-propelled and towed vehicles.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements. if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Command, U.S. Army ARDEC, ATTN : SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b. Alternative quality conformance Provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.5 Subject term (key word) listing.

Fire Control System (FCS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A623)

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